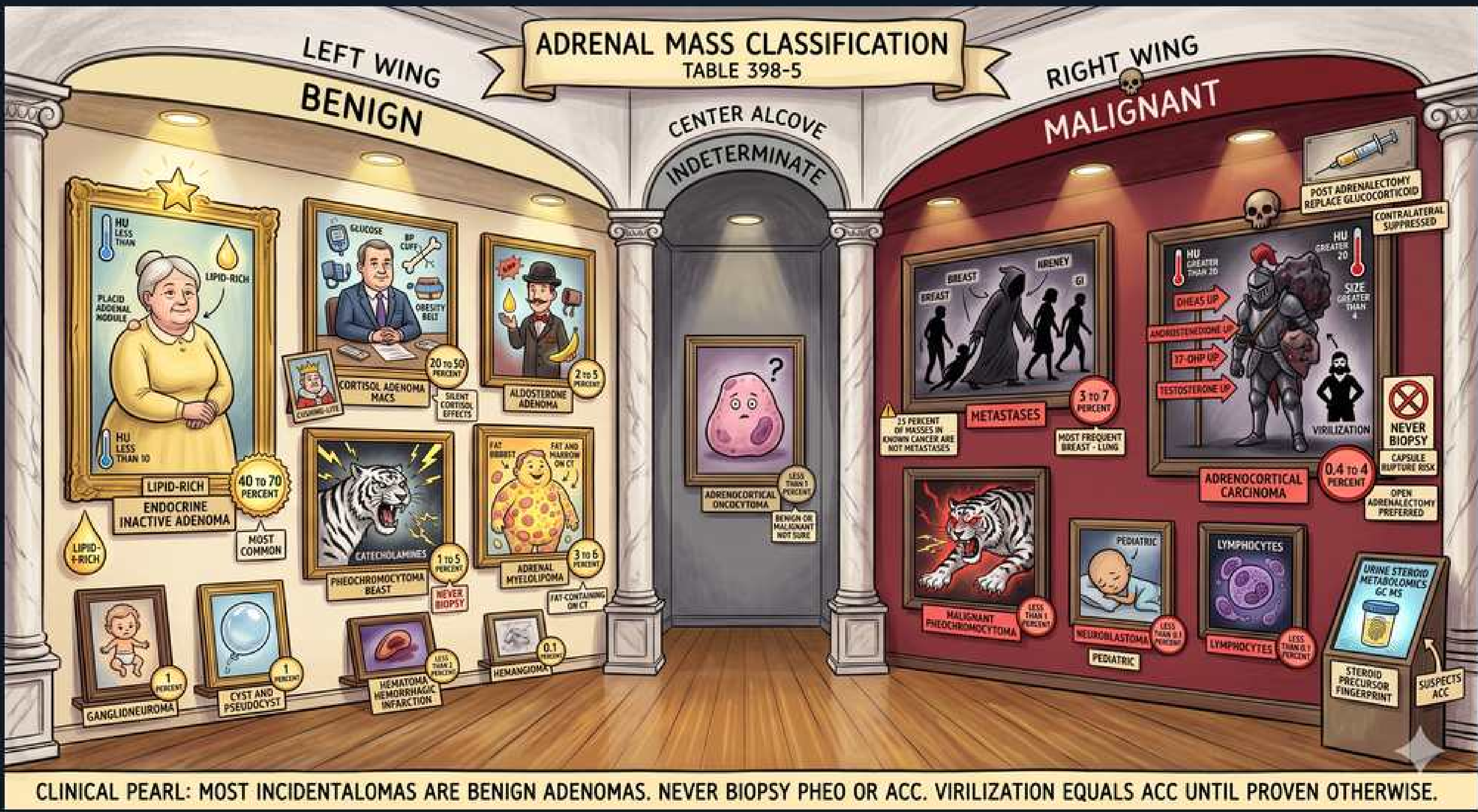
Adrenal — Adrenal Mass Gallery (Incidentaloma)



1. Adrenal mass classification

WHAT YOU SEE

Title banner 'ADRENAL MASS CLASSIFICATION TABLE 398-5' over a museum-style gallery — the whole scene is a gallery sorting every adrenal mass into benign, indeterminate, and malignant wings.

WHAT IT ENCODES

An adrenal incidentaloma is a mass ≥1 cm found on imaging done for another reason; prevalence rises with age (~4–7% on CT). Workup answers two parallel questions — is it FUNCTIONAL (hormone screen) and is it MALIGNANT (imaging phenotype + size) — and sorts lesions into benign, indeterminate, and malignant.

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2. Benign left wing

WHAT YOU SEE

Left wing of the gallery under a warm-lit arch reading 'BENIGN', hung with framed portraits of harmless lesions — the cozy, sunny wing signals the reassuring, no-further-workup masses.

WHAT IT ENCODES

The vast majority of incidentalomas are BENIGN, non-functioning cortical adenomas. Benign features: small (<4 cm), homogeneous, smooth margins, non-contrast ≤10 HU (lipid-rich), and rapid contrast washout (absolute >60% / relative >40%) for lipid-poor ones.

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3. Lipid-rich adenoma (≤10 HU)

WHAT YOU SEE

Picture 'LIPID-RICH ENDOCRINE INACTIVE ADENOMA, ≤10 HU'

WHAT IT ENCODES

The lipid-rich adenoma is the prototypical benign incidentaloma: intracytoplasmic fat drops non-contrast attenuation to ≤10 HU, which is highly specific for a benign adenoma — no biopsy and, if non-functional and <4 cm, generally no imaging follow-up. Lipid-POOR adenomas read 10–20 HU and are confirmed benign by washout or MRI chemical shift (signal drop on out-of-phase).

BOARD TRAP

≤10 HU is reassuring and does NOT need biopsy or even routine repeat imaging if non-functional — over-biopsying a clear lipid-rich adenoma is the trap.

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4. Myelolipoma

WHAT YOU SEE

Framed portrait of a yellow fat-and-marrow lesion labelled 'MYELOLIPOMA' — the bright fat-yellow drawing IS the macroscopic fat (very negative HU) that pathognomonically marks it benign.

WHAT IT ENCODES

Myelolipoma is a benign tumor of mature adipose tissue plus hematopoietic (myeloid) elements; the pathognomonic finding is MACROSCOPIC FAT (very negative HU, like subcutaneous fat) on CT/MRI. It is non-functioning and needs no workup; only large ones (>4 cm, risk of hemorrhage) are considered for resection.

BOARD TRAP

Macroscopic fat = myelolipoma (benign) — don't confuse the negative-HU fat of myelolipoma with the low-but-positive ≤10 HU of a lipid-rich adenoma.

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5. Cyst / hemorrhage / ganglioneuroma

WHAT YOU SEE

Benign cyst, hemorrhage, ganglioneuroma thumbnails

WHAT IT ENCODES

Other clearly benign entities: a simple ADRENAL CYST (water density, thin wall), a HEMORRHAGE/hematoma (often post-trauma/anticoagulation, density evolves and resolves over time), and a GANGLIONEUROMA (well-differentiated benign neural-crest tumor). These have characteristic imaging and need no hormonal/oncologic chase once typed.

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6. Indeterminate center alcove

WHAT YOU SEE

Pink mass with '?' in central alcove 'INDETERMINATE'

WHAT IT ENCODES

Indeterminate lesions are those not meeting benign criteria — typically 10–20+ HU, lipid-poor, or with incomplete washout. Next steps: CT washout study, MRI chemical-shift, FDG-PET, or short-interval imaging follow-up (e.g. repeat in 6–12 months); persistent suspicion or growth → resection/MDT.

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7. Malignant right wing

WHAT YOU SEE

Right wing under a blood-red arch reading 'MALIGNANT', hung with sinister portraits — the dark red wing signals the aggressive, must-not-miss masses.

WHAT IT ENCODES

Malignant phenotype: large (>4 cm, especially >6 cm), heterogeneous, >20 HU non-contrast, irregular margins, calcification/necrosis, slow washout, local/vascular invasion, FDG-avid. Categories include metastases, ACC, and pheochromocytoma (which behaves as a surgical 'must-not-miss').

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8. Metastases

WHAT YOU SEE

Silhouette family 'METASTASES — most frequent malignancy'

WHAT IT ENCODES

Metastases are the MOST COMMON malignant adrenal lesion overall — primaries include lung, breast, melanoma, renal, and GI cancers; often bilateral. In a patient with known cancer, a new indeterminate adrenal mass is statistically a metastasis, and biopsy may be appropriate to confirm — but ONLY after biochemically excluding pheo.

BOARD TRAP

Don't assume a solitary adrenal mass in a cancer patient is automatically a metastasis without characterization, and never biopsy it before excluding pheo.

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9. Adrenocortical carcinoma >4cm

WHAT YOU SEE

Muscular villain 'ADRENOCORTICAL CARCINOMA, >4cm', >20 HU

WHAT IT ENCODES

ACC is a rare, aggressive primary: typically large (>4–6 cm), heterogeneous, >20 HU, irregular, often hormone-secreting (cortisol ± androgens → Cushing + virilization). Management is complete OPEN resection (R0) to avoid capsule rupture, with adjuvant mitotane for high-risk disease; never biopsy.

BOARD TRAP

A rapidly growing, large, heterogeneous mass with mixed hormone excess is ACC — resect (open), don't biopsy or 'watch.'

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10. Pheochromocytoma (tiger)

WHAT YOU SEE

Tiger image (pheo) in malignant wing

WHAT IT ENCODES

Pheochromocytoma (the 'tiger') is a catecholamine-secreting tumor — classically markedly T2-bright on MRI, often >20 HU and very vascular, ~10% bilateral/familial (MEN2, VHL, NF1, SDH). ALWAYS screen with plasma free / 24-h urine fractionated metanephrines before touching any adrenal mass; if positive, alpha-blockade (phenoxybenzamine) before beta-blockade, then surgery.

BOARD TRAP

Biopsy or unprepped surgery on a pheo can trigger a fatal hypertensive crisis — screen for and exclude pheo first, alpha-block before beta-block.

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11. Never biopsy

WHAT YOU SEE

Red 'NEVER BIOPSY' sign

WHAT IT ENCODES

Biopsy is contraindicated for suspected ACC (cannot diagnose without the whole specimen + Weiss score; risks needle-track seeding and capsule violation) and for pheo (crisis). Biopsy is reserved for a probable metastasis in a known-cancer patient, performed only AFTER pheo exclusion.

BOARD TRAP

Biopsy is contraindicated for pheo and suspected ACC — the only legitimate biopsy is a suspected metastasis after pheo is excluded.

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12. Lymphoma / unusual malignancy

WHAT YOU SEE

Cellular thumbnail 'LYMPHOMA' in malignant wing

WHAT IT ENCODES

Primary adrenal lymphoma is rare, usually non-Hodgkin (DLBCL), frequently BILATERAL and can cause adrenal insufficiency by destroying both glands; secondary adrenal involvement by systemic lymphoma is more common. Unlike ACC, this is a setting where biopsy IS appropriate for diagnosis (after pheo exclusion), as treatment is chemotherapy, not surgery.

BOARD TRAP

Bilateral adrenal masses with adrenal insufficiency suggest lymphoma, metastases, infection (TB), or hemorrhage — not a solitary functioning adenoma.

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13. Clinical pearl

WHAT YOU SEE

'MOST INCIDENTALOMAS ARE BENIGN ADENOMAS. NEVER BIOPSY PHEO OR ACC. VIRILIZATION EQUALS ACC UNTIL PROVEN OTHERWISE'

WHAT IT ENCODES

Bottom line: most incidentalomas are benign non-functioning adenomas; never biopsy a pheo or suspected ACC; new virilization (or rapidly mixed Cushing) with an adrenal mass means ACC until proven otherwise. Screen every mass for pheo, Cushing, and (if HTN/hypokalemia) aldosteronism, and resect if >4 cm, functional, or imaging-suspicious.
